

Subgame Perfection and Win-Lose Games

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Subgame Perfect Equilibrium

- Last time we solved sequential games using **backward induction**.
- Today we refine Nash equilibrium using this idea: the **subgame perfect equilibrium**.
- This lets us identify which threats are credible and which are empty.
- We'll also look at **win-lose games**, where backward induction yields clean results.

Associated Reading

Game Theory, Sections 3.4 and 3.5

Recap: The Entry Deterrence Game

Entrant

├ Stay Out $\rightarrow (0, 4)$

└ Enter

└ Incumbent

├ Fight $\rightarrow (-1, 1)$

└ Accommodate $\rightarrow (2, 2)$

Payoffs: (Entrant, Incumbent)

By backward induction: the Incumbent would Accommodate ($2 > 1$), so the Entrant Enters ($2 > 0$). Outcome: (Enter, Accommodate) with payoffs (2, 2).

Why the Incumbent's Threat Fails

The Incumbent might *prefer* the Entrant to Stay Out (they get 4 instead of 2).

Could the Incumbent **threaten** to Fight if the Entrant enters?

The threat would require fighting, which gives the Incumbent payoff 1. But a rational Incumbent prefers 2 (Accommodate).

The threat is **not credible**. A rational Entrant knows the Incumbent won't carry it out.

This is one of the main insights of backward induction: it reveals which threats and promises are credible.

Subgame Perfect Equilibrium

What Is a Subgame?

A **subgame** is any part of the game tree that:

- Begins at a single decision node.
- Contains all branches following from that node.
- Is a complete game in itself.

In the entry game: the whole game is a subgame, and so is the Incumbent's decision node after entry.

A **subgame perfect equilibrium** (SPE) is a strategy profile that constitutes a Nash equilibrium **in every subgame**.

This rules out Nash equilibria that rely on **non-credible threats or promises** off the equilibrium path.

In perfect information games: the backward induction outcome is always the unique SPE.

A Nash Equilibrium that isn't Subgame Perfect

In the entry game, consider the strategy profile:

- Entrant: **Stay Out**
- Incumbent: **Fight** (if entered)

Is this a Nash equilibrium?

Yes. Given the Incumbent will Fight, the Entrant's best response is to Stay Out ($0 > -1$). And the Incumbent's strategy doesn't matter since the node is not reached.

But it's **not subgame perfect**. At the Incumbent's subgame, Fight is not a best response (Accommodate gives $2 > 1$).

The Importance of Subgame Perfection

Subgame perfection refines Nash equilibrium by requiring rationality **everywhere** in the game tree, not just on the equilibrium path.

It eliminates equilibria held in place by empty threats.

This makes it the dominant solution concept for perfect information sequential games.

More Examples

The Centipede Game

Two players alternate between *continuing* (C) or *stopping* (S) the game. Each stage, the pot grows, but the player who stops takes more.

P1 P2 P1
C → C → C → C → C → S
(or stop at any point)

At each node, stopping gives the current player a larger share than continuing, *if* the other player is about to stop.

Backward Induction in Centipede

Apply backward induction from the end:

- At the last node: Player 1 stops (gets more).
- At the second-to-last node: Player 2 knows P1 will stop, so P2 stops.
- At each earlier node: same logic applies.
- **Result:** Player 1 stops immediately, at the very first move.

The backward induction prediction is that no cooperation ever happens, even though both players would do much better if they cooperated longer.

The Paradox of Backward Induction

The Centipede Game shows a tension:

- **Backward induction** predicts immediate stopping.
- But this requires Player 1 to reason: “If I reach move 3, Player 2 is rational... but how did we reach move 3?”

In experiments, people rarely stop immediately. Cooperation often unfolds for many rounds.

Whether this shows backward induction is *wrong*, or merely that people are *not* perfectly rational, is a substantive debate.

Win-Lose Games

Consider two-player perfect-information games where the only outcomes are “Player 1 wins” or “Player 2 wins.” We assign payoffs $(1, 0)$ and $(0, 1)$ respectively.

These **win-lose games** have a remarkably clean structure when analyzed by backward induction.

An Example: The Number Game

Two players take turns choosing a number from $\{1, 2, \dots, 10\}$. Player 1 goes first. The player whose choice brings the running total to 100 or more wins.

Player 1 has a winning strategy. By backward induction, the “losing positions” (where the player whose turn it is cannot win) are: 89, 78, 67, 56, 45, 34, 23, 12, 1.

Player 1 starts with 1. Then, whatever Player 2 picks (call it n), Player 1 picks $11 - n$. This brings the total to the next losing position for Player 2 each time.

Theorem (Zermelo, 1913): In every finite two-player win-lose game with perfect information, one of the two players has a **winning strategy**.

The proof uses backward induction. At the root, either some branch leads to a $(1, 0)$ outcome (Player 1 can win), or all branches lead to $(0, 1)$ outcomes (Player 2 wins regardless).

What About Draws?

Games like chess and checkers allow three outcomes: Player 1 wins, Player 2 wins, or Draw.

An extension of Zermelo's theorem says exactly one of three cases holds:

- Player 1 has a strategy guaranteeing a win.
- Player 2 has a strategy guaranteeing a win.
- Both players have strategies guaranteeing at least a draw, so with optimal play the outcome is a draw.

Checkers falls into the third category (proven by computer in 2007). For chess, we still do not know which category applies.

Worked Example

A Regulation Game

A company (Player 1) decides whether to **Pollute** (cheaper production) or go **Clean**. If the company pollutes, a regulator (Player 2) can **Fine** the company or **Ignore** the violation.

Company



Payoffs: (Company, Regulator)

Part 1: Backward Induction

Step 1: At the Regulator's node, Fine gives 1 and Ignore gives 0. The Regulator chooses **Fine**.

Step 2: The Company anticipates this. Clean gives 2; Pollute gives 0 (since the Regulator will Fine). The Company chooses **Clean**.

Backward induction outcome: Clean, with payoffs (2, 3).

Part 2: A Non-Credible Threat?

Suppose the Regulator announces: “We will Ignore any pollution.”

If the Company believed this, they would Pollute (payoff 4 vs. 2). The outcome would be (Pollute, Ignore) with payoffs (4, 0).

Is (Pollute, Ignore) a Nash equilibrium?

Check: Given Ignore, the Company prefers Pollute ($4 > 2$). Given Pollute, does the Regulator prefer Ignore? No, Fine gives $1 > 0$. So this is **not** a Nash equilibrium. The Regulator's promise to ignore violations is not even self-consistent.

Part 3: What About (Clean; Fine)?

Consider: Company plays Clean, Regulator plays Fine.

Given the Regulator will Fine, Company prefers Clean ($2 > 0$). Given the Company plays Clean, the Regulator's strategy doesn't affect the outcome (the Fine node is never reached). So the Regulator has no reason to deviate.

This **is** a Nash equilibrium, and it's also subgame perfect. In the subgame starting at the Regulator's node, Fine is the best response ($1 > 0$).

Part 4: Could (Clean; Ignore) Be a Nash Equilibrium?

Given the Regulator plays Ignore, Company prefers Pollute ($4 > 2$). So Company would deviate.

This is **not** a Nash equilibrium. The Regulator's lax policy only works as a strategy if the Company never actually tests it, but the Company *would* test it.

This example shows why backward induction and subgame perfection matter: they force us to check what happens at decision nodes that are “off the equilibrium path.”

- We move to games where players may **randomise** by choosing strategies with probability.
- This requires cardinal payoffs (not just ordinal rankings).
- And it opens up a new kind of Nash equilibrium: **mixed strategy equilibrium**.